Proof checks: all attempts retained

27 October 2026. Lean4.34.1 checks eight final theorems. The unchanged repository axiom auditor reports eight checked, zero failing and zero trusted, with empty axiom lists. Four literal vector inequalities and the generic monotone-budget implication are proved. Monotonicity and transitive comparison are explicit assumptions; this does not prove anything about unrecorded historical runtime, the original caller, curve security or an attack. Native literal-data binding matches the independently checked table.

Proof development; final kernel and axiom audit pass

Attempt	Exit status	Recorded outcome
001	build 1 / audit —	Thread startup failed under4GiB virtual-address cap; default workers
002	build 1 / audit —	Same startup failure with one worker and4GiB virtual-address cap
003	build 1 / audit —	1GiB Lean allocation limit, no virtual-address cap; CountsLE needed unfolding for decidability; error placeholders not accepted as proofs
004	build 0 / audit 1	Six kernel proofs pass; auditor import exceeds1GiB allocation budget
005	build 0 / audit 1	Final eight kernel proofs pass;4GiB auditor cannot locate its external lean command on PATH
006	build — / audit 0	Verified binary added to command-local PATH; unchanged repository auditor checks all8 with0axioms and0trusted

Statuses only; final eight kernel proofs and unchanged axiom auditor pass · no measured clock

The diagram preserves sequence and status instead of comparing cost: no runtime or RSS was measured. Attempts001 and002 fail thread startup under a 4 GiB virtual-address cap. Attempt 003 fails decidability elaboration; unfolding the unchanged predicate resolves it without changing vectors or statements. Attempt 004 builds six proofs but its1GiB auditor exceeds its allocation. Attempt 005 builds the final eight, while its auditor cannot locate external lean on PATH. Audit 006 passes with the verified binary on command-local PATH. Failed outputs and source snapshots are preserved, not accepted as proofs.

Final kernel checking uses one worker and1GiB internal allocation limit; auditing uses one worker and4GiB, with no virtual-address cap or measured RSS. Shared local host, no isolation tier, timing reference, random seed or legacy execution. The official release bytes and published SHA match. lean4checker, elan and lake are unavailable; exact compiler and auditor commands are retained. This is not a full repository proof-gate or hosted CI pass. These formalization attempts are additions after validation, not the frozen primary source revision. Their source hashes and commands are in the evidence bundle.

Proof map, attempts, commands, kernel output, auditor output, reproduction. Raw archive paths runs/proof-attempts/build001 through build005 and audit006 retain every failure.